Exact Spectra for Hyperplane Chamber Walks: A Route-A Stress Test

HCS-C192 theorem and evidence package

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Abstract

For every finite real hyperplane arrangement and every probability measure on its faces, the Brown–Diaconis theorem diagonalizes the induced chamber walk. We record the resulting characteristic polynomial, finite determinant, power traces, unique-stationarity criterion, exact sampler, and mixing bound. Exact rational coordinate and braid fixtures test the formulas but do not prove the all-family theorem. The complete operator structure still fails Route A: without target arithmetic semantics, its determinant is only a formal hint.

1 Face dynamics and the flat spectrum

Let $\mathcal{A}$ be a finite real hyperplane arrangement in V, with faces $\mathcal{F}$ and chambers $\mathcal{C}$. In sign coordinates the face product is $(FG)_H = F_H$ when $F_H \neq 0$ and $(FG)_H = G_H$ otherwise. A face law w defines

$$K(C, C') = \sum_{F : FC = C'} w(F).$$

Write $L(\mathcal{A})$ for the intersection poset and μ for its Möbius function. Brown and Diaconis [1, Theorem 1] prove that K is diagonalizable, with flat-indexed data

$$\lambda_W = \sum_{F \subseteq W} w(F), \qquad m_W = |\mu(W, V)|.$$

Consequently, even when several flats yield the same numerical eigenvalue,

$$\chi_K(x) = \prod_{W \in L(\mathcal{A})} (x - \lambda_W)^{m_W},$$

$$\det(I - zK) = \prod_{W \in L(\mathcal{A})} (1 - z\lambda_W)^{m_W},$$

$$\operatorname{tr}(K^\ell) = \sum_W m_W \lambda_W^\ell.$$

These are finite-dimensional deductions from the cited theorem, not a new diagonalization claim. In particular, diagonalizable does not mean reversible or self-adjoint.

2 Stationarity and convergence

A face law is *separating* if each $H \in \mathcal{A}$ is avoided by some positive-weight face. Brown–Diaconis Theorem 2 proves that separation is equivalent to unique stationarity. Under separation, sample all positive-weight faces without replacement, choosing each next face proportionally to its remaining weight. Their product is a chamber with the stationary law π.

For every start C and $\ell \geq 0$, their coupling argument gives

$$\|K_C^\ell - \pi\|_{\mathrm{TV}} \leq -\sum_{W \neq V} \mu(W, V)\lambda_W^\ell \leq \sum_{H \in \mathcal{A}} \lambda_H^\ell.$$

The middle term is exactly the probability that the product of ℓ i.i.d. sampled faces is not yet a chamber [1, Section 4B].

3 Finite exact regression

Two independent rational implementations and a separate SymPy oracle check eight coordinate/braid fixtures. The census contains 316 faces, 94 chambers, 75 flats, 1,604 transition cells, 62 stationary probabilities, 24 mixing rows, and 48 trace rows. The independent checker passes 20,609 assertions; SymPy passes 3,398 exact checks. These finite examples are regression oracles only.

4 Route-A verdict

The exact result has no intrinsic rational-prime or target-zero index, recovers no target arithmetic data, supplies no target functional equation, and yields no target counting law. Its finite determinant is a natural operator object but no target divisor is identified. Hence

$$(A0, A1, A2, A3, A4) = (\mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FORMAL\_HINT}),$$

with overall rejection and no Route-B invocation. No local or Euler factors, root numbers, automorphy, Hilbert–Pólya operator, global novelty, or external review are claimed.

References

[1] K. S. Brown and P. Diaconis, Random walk and hyperplane arrangements, *Ann. Probab.* 26 (1998), 1813–1854, doi:10.1214/aop/1022855884.